THE COHEN-MACAULAY LOCUS OF A FLAT FINITELY PRESENTED MAP OF SPACES IS OPEN AND BASE-CHANGE STABLE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 6402-6416 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 6418-6433, the complete author proof (14 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Counted B side (spaces, more general; both sides complete). Companion chart-import shape to wave 14's CM family.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Definition of Cohen-Macaulay morphism of spaces via charts (in-source); More on Morphisms, lemma flat-finite-presentation-CM-open.

Source reference. Original lemma 045U: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 6433–6433. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S which is flat and locally of finite presentation. Let*

$$W = \{x \in |X| : f \text{ is Cohen-Macaulay at } x\}$$

Then W is open in $|X|$ and the formation of W commutes with arbitrary base change of f: For any morphism $g : Y' \to Y$, consider the base change $f' : X' \to Y'$ of f and the projection $g' : X' \to X$. Then the corresponding set W' for the morphism f' is equal to $W' = (g')^{-1}(W)$.

Proof. Choose a commutative diagram

$$\begin{array}{ccc} U & \longrightarrow & V \\ \downarrow & & \downarrow \\ X & \longrightarrow & Y \end{array}$$

with étale vertical arrows and U and V schemes. Let $u \in U$ with image $x \in |X|$. Then f is Cohen-Macaulay at x if and only if $U \to V$ is Cohen-Macaulay at u (by

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definition). Thus we reduce to the case of the morphism $U \to V$. See More on Morphisms, Lemma 045U. $\square$